

석사학위논문
Master's Thesis

가중치 통합과 연속적 학습을 활용한 오픈소스
평가 언어 모델의 신뢰성 향상 연구

Building Reliable Open-Source Language Model Evaluators
through Weight Merging and Continual Training

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The study was conducted in accordance with Code of Research Ethics¹.

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MAI

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초 록

본 논문은 LLM 평가의 신뢰성과 투명성을 높이기 위해 가중치 병합 기반 통합 평가자 모델 Prometheus 2를 제안한다. 기존 오픈소스 평가자는 인간·GPT-4와의 상관성이 낮고 절대·쌍대 평가를 모두 처리하기 어려웠다. Prometheus 2는 두 형식에 특화된 모델을 각각 학습한 뒤 병합하여 이 한계를 해결했다. 1천 개의 평가 기준과 20만 개의 쌍대 비교 데이터를 활용해 학습된 Prometheus 2는 Vicuna Bench, MT Bench 등에서 인간·GPT-4와 0.6 이상의 상관을 달성하며 기존 대비 약 0.2포인트 향상된 성능을 보였다. 또한 쌍대 비교에서는 인간 판단과의 일치율이 두 배가량 증가했다. BiGGen-Bench 기반 지속 학습으로 개발한 Prometheus-2-BGB는 인간 상관이 0.471에서 0.577로, GPT-4 상관이 0.688에서 0.836으로 향상되었으며 자기 일관성 디코딩 사용 시 인간 상관은 0.607까지 증가했다. 다른 벤치마크 성능도 유지되어 치명적 망각 없이 특정 평가에 적응 가능성을 확인했다.

핵심 낱말 LLM 평가, 오픈소스 평가자, 언어 모델 평가 신뢰성, 평가 투명성, 절대 평가, 쌍대 비교, 가중치 병합, 지속 학습, 인간 상관성, 모델 일반화

Abstract

This paper introduces Prometheus 2, a unified evaluator model designed to enhance the reliability and transparency of LLM evaluation. Existing open-source evaluators show low correlation with human and GPT-4 judgments and struggle to handle both direct assessment and pairwise comparison. Prometheus 2 overcomes these limitations by separately training models for each evaluation format and merging their weights. Trained on 1,000 fine-grained criteria and 200,000 pairwise comparisons, Prometheus 2 achieves correlations above 0.6 with human and GPT-4 evaluators on benchmarks such as Vicuna Bench and MT Bench, improving by about 0.2 points over prior open-source models. In pairwise settings, agreement with human judgments nearly doubles. Through continual training with BiGGen-Bench data, Prometheus-2-BGB improves human correlation from 0.471 to 0.577 and GPT-4 correlation from 0.688 to 0.836, reaching 0.607 with self-consistency decoding. Crucially, it adapts to specific benchmarks without catastrophic forgetting.

Keywords LLM evaluation, open-source evaluator, evaluation reliability, evaluation transparency, direct assessment, pairwise comparison, weight merging, continual learning, human correlation, model generalization

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Chapter 1. Introduction

Evaluating the quality of outputs produced by large language models (LLMs) has become increasingly challenging as models are applied to an ever-expanding range of complex, open-ended tasks. Traditional automatic metrics such as BLEU [Papineni et al., 2002] and ROUGE [Lin, 2004] capture surface-level similarity but fail to reflect nuanced qualities such as reasoning depth, factual accuracy, or stylistic appropriateness. Human evaluation remains the gold standard, yet it is costly, time-consuming, and difficult to scale.

To mitigate these issues, language-model-based evaluation has emerged as a scalable and flexible paradigm [Li et al., 2024, Gao et al., 2024]. In this setting, an evaluator LM is prompted to judge the quality of a model’s response either by producing a scalar score (direct assessment) [Zheng et al., 2023, Liu et al., 2023b, Ye et al., 2023, Kim et al., 2023b] or by selecting the preferred output among two candidates (pairwise ranking) [Wang et al., 2023b, Li et al., 2023b, Lambert et al., 2024]. Proprietary LMs such as GPT-4 or Claude-3 have demonstrated high correlation with human judgments under this paradigm [Dubois et al., 2023, Ye et al., 2023], showing that LMs can serve as effective judges.

However, relying on proprietary evaluators introduces several critical challenges. Because their training data and update schedules are undisclosed, reproducibility and fairness cannot be guaranteed [Kim et al., 2023b]. Moreover, their usage entails high financial cost and limited controllability, both of which restrict practical and academic adoption. Consequently, recent work has focused on developing open-source evaluator LMs [Kim et al., 2023b, Wang et al., 2023b, Li et al., 2023a, Zhu et al., 2023, Jiang et al., 2023b, Lee et al., 2024]. While these models improve transparency, they often yield scores that correlate weakly with human or GPT-4 evaluations and are typically trained for only one evaluation format—either direct assessment or pairwise ranking—making them inflexible in real-world use.

To bridge this gap, we propose Prometheus 2, an open-source evaluator LM that unifies both evaluation paradigms through weight merging. Our approach merges two separately trained evaluator LMs—one trained on direct-assessment feedback (the Feedback Collection [Kim et al., 2023b]) and the other on pairwise-ranking feedback (the newly introduced Preference Collection)—to form a single robust evaluator. This merging strategy produces an evaluator that not only performs both tasks seamlessly but also surpasses jointly trained or single-format baselines in correlation with human and proprietary judges. Trained on 1,000 fine-grained evaluation rubrics and 200,000 pairwise comparison examples, Prometheus 2 (7B and 8×7B) achieves over 0.6 Pearson correlation with human and GPT-4 evaluators on direct-assessment benchmarks (Vicuna Bench, MT Bench, FLASK, Feedback Bench) and halves the performance gap with GPT-4 on pairwise-ranking benchmarks (HHH Alignment, MT Bench Human Judgment, Auto-J Eval, Preference Bench).

While Prometheus 2 demonstrates strong general-purpose evaluation capabilities, deploying evaluators in real-world scenarios presents additional practical challenges. LLM developers often maintain internal benchmarks specific to their domains or use cases, and relying on proprietary evaluators for these custom benchmarks incurs ongoing costs that scale with evaluation frequency. Fine-tuning proprietary evaluators on custom feedback is prohibitively expensive, and developing benchmark-specific evaluators from scratch requires substantial resources. This motivates an important research question: can we adapt a general-purpose open-source evaluator to specific benchmarks through continual training on benchmark-specific feedback, while maintaining performance on other evaluation tasks?

To address this question, we investigate continual training methodology for evaluator adaptation. Using BiGGen-Bench (a fine-grained generative benchmark spanning diverse capabilities) as a validation case study, we continually train Prometheus 2 on benchmark-specific evaluation feedback to create Prometheus-2-BGB. This approach combines two complementary improvement strategies: (1) continual feedback training on BiGGen-Bench-specific rubrics and model outputs, and (2) self-consistency decoding that aggregates multiple evaluation samples to reduce variance. The resulting specialized evaluator demonstrates whether open-source evaluators can be efficiently adapted to specific evaluation contexts without sacrificing their general-purpose capabilities.

Empirically, Prometheus-2-BGB achieves substantial performance improvements on BiGGen-Bench evaluation tasks. The base Prometheus 2 model shows 0.471 Pearson correlation with human judgments, while Prometheus-2-BGB improves this to 0.577—a 22% relative improvement. When combined with self-consistency decoding ($N=3$ samples), correlation further increases to 0.607. Importantly, Prometheus-2-BGB also achieves 0.836 correlation with GPT-4 evaluations (compared to 0.688 for base Prometheus 2), effectively learning to mimic GPT-4’s evaluation behavior at a fraction of the cost. Critically, continual training exhibits no catastrophic forgetting: Prometheus-2-BGB maintains performance on other standard benchmarks (FLASK, Feedback Bench, MT Bench), demonstrating positive task transfer rather than interference. These results suggest that continual training provides a practical pathway for organizations to adapt open-source evaluators to their specific needs while preserving general evaluation capabilities.

In summary, this thesis advances open-source LLM evaluation through two primary contributions: (1) Prometheus 2, a unified evaluator LM that integrates direct assessment and pairwise ranking through weight merging, achieving human-level reliability across diverse benchmarks; and (2) a continual training methodology that enables efficient adaptation of evaluators to specific benchmarks, demonstrated through Prometheus-2-BGB, which achieves near-GPT-4-level performance on specialized evaluation tasks without catastrophic forgetting. Together, these contributions establish a foundation for transparent, affordable, and adaptable LLM evaluation pipelines that can serve both research and industry needs.

The remainder of this thesis is organized as follows. Chapter 2 reviews related work on language model-based evaluation, weight merging techniques, and continual learning approaches. Chapter 3 presents Prometheus 2, detailing the weight merging methodology, training data construction, and experimental validation across multiple benchmarks. Chapter 4 investigates continual training for benchmark-specific adaptation, demonstrating how Prometheus 2 can be efficiently adapted to BiGGen-Bench while maintaining general evaluation capabilities. Chapter 5 concludes with a discussion of limitations and promising directions for future work.

1.1 Related Work

1.1.1 Language Model-based Evaluation

To assess the generation capabilities of LMs, prior works such as the GEM benchmark [Gehrmann et al., 2021, 2022] employed ROUGE [Lin, 2004], BLEU [Papineni et al., 2002], and BERTScore [Zhang et al., 2019] as their metrics, which measure the lexical or semantic similarity between a reference answer and a response. However, these conventional metrics are prone to false negatives because they are not expressive enough to recognize responses that are of good quality but differ from the reference

answer [Schluter, 2017, Freitag et al., 2020, Hanna and Bojar, 2021].

Recently, employing language models as a judge has gained attention as a promising paradigm to mimic the depth and granularity that human evaluation offers [Zheng et al., 2023, Liu et al., 2023b, Li et al., 2023b, Chan et al., 2023, Ye et al., 2023]. To reduce the over-reliance on proprietary LMs, follow-up works suggest training language models specialized in evaluations [Cui et al., 2023, Kim et al., 2023b, Jiang et al., 2023b,c, Li et al., 2023a, Lee et al., 2024]. Yet, open evaluator LMs do not possess the flexibility to function in different evaluation schemes and show weak evaluation performance compared to proprietary LMs. Recent work has explored weight merging approaches to create unified evaluators that can operate across multiple evaluation formats while maintaining competitive performance.

1.1.2 Weight Merging

Prior works have demonstrated that weight merging can enhance performance across various domains, including language modeling [Li et al., 2022, Matena and Raffel, 2022, Ilharco et al., 2022, Don-Yehiya et al., 2022, Gururangan et al., 2023, Yadav et al., 2024, Sukhbaatar et al., 2024], instruction-tuning [Jang et al., 2023b, Yu et al., 2023], and aligning to user preferences [Jang et al., 2023a, Rame et al., 2024, Wang et al., 2024]. Recent work has applied weight merging to enhance the evaluation capabilities of open evaluator LMs by merging models trained on different assessment formats—specifically, direct assessment and pairwise ranking—to obtain unified evaluators that function in both formats while achieving performance comparable to proprietary LMs.

1.1.3 Continual Learning and Domain Adaptation

Continual learning allows models to adapt to new tasks while retaining performance on previously learned ones [Kirkpatrick et al., 2017, McCloskey and Cohen, 1989]. A key challenge in this paradigm is catastrophic forgetting, where models lose previously acquired knowledge when trained on new data. In the context of evaluator LMs, this raises an important practical question: can we adapt a general-purpose evaluator to specific benchmarks by training on feedback from that benchmark, thereby creating an affordable internal evaluation pipeline? Prior work has shown that fine-tuning on domain-specific data can improve performance [Scialom et al., 2022], but the risk of catastrophic forgetting must be carefully monitored. Continual training approaches have been explored to address this challenge in various domains, providing a foundation for adapting evaluator models to specific benchmarks while preserving their general capabilities.

Chapter 2. Prometheus 2: Unified Evaluator via Weight Merging

2.1 Motivation

Evaluating the quality of outputs produced by language models (LMs) is progressively becoming difficult, as the outputs cover an extremely diverse distribution of text and complex tasks. To address this issue, language model-based evaluation has emerged as a scalable and affordable paradigm for assessing LM-generated text [Li et al., 2024, Gao et al., 2024]. In this paradigm, LMs are either prompted to output a scalar indicator of quality (denoted as *direct assessment*) [Zheng et al., 2023, Liu et al., 2023b, Ye et al., 2023, Kim et al., 2023b] or to determine which of two outputs are preferred (denoted as *pairwise ranking*) [Wang et al., 2023b, Li et al., 2023b, Lambert et al., 2024]. Prior works employing proprietary LMs as evaluators have demonstrated not only high correlations with human evaluations but also increased speed and cost-effectiveness [Zheng et al., 2023, Liu et al., 2023b, Dubois et al., 2023, Ye et al., 2023].

However, relying on proprietary LMs for evaluation poses significant challenges. The lack of transparency about their training data compromises both fairness and reproducibility, making it problematic to use them in evaluation pipelines. Additionally, concerns regarding controllability and affordability persist [Kim et al., 2023b]. To address these issues, recent works have focused on developing evaluator LMs that are open-access, transparent, and controllable [Kim et al., 2023b, Wang et al., 2023a,b, Li et al., 2023a, Zhu et al., 2023, Jiang et al., 2023b,c, Lee et al., 2024]. Yet, these models often yield scoring decisions that do not correlate well enough with human judgments or those made by proprietary LMs, failing to effectively simulate them. Moreover, open evaluator LMs are not flexible since they are typically trained only to perform either direct assessment or pairwise ranking and assess based on general public preferences like helpfulness and harmlessness, limiting their ability to handle diverse real-life scenarios.

To close the gap with proprietary LMs, we investigate *unifying* the two model-based evaluation paradigms—direct assessment and pairwise ranking—to train a robust unified evaluator LM. We propose a recipe based on merging the weights of two evaluator LMs trained separately on direct assessment and pairwise ranking formats. Our key empirical observation is that weight merging can yield an evaluator LM that not only *works* in both formats, but also *outperforms* evaluator LMs that are jointly trained or only trained on a single format. As shown in Figure 2.1, weak evaluators such as Llama-2-Chat-70B, Prometheus, and GPT-3.5-Turbo achieve low scoring correlation with strong evaluators like humans, GPT-4, and Claude-3-Opus. In contrast, scores provided by strong evaluators highly correlate with each other, demonstrating the importance of developing strong open-source evaluators.

2.2 Evaluation Formats

Language model-based evaluation can be performed in two primary formats: direct assessment and pairwise ranking. Both evaluation schemes play crucial roles in different evaluation scenarios, and a unified evaluator must excel in both formats to be practically useful. Figure 2.2 illustrates the difference

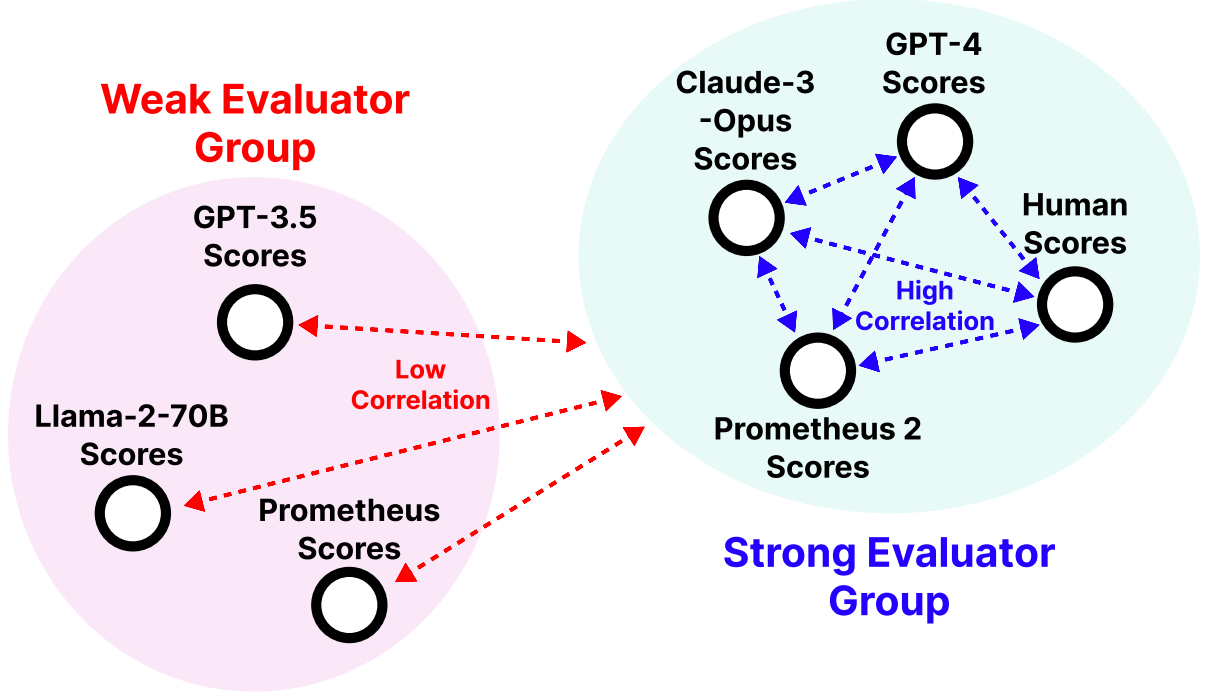


Figure 2.1: Weak evaluators (*e.g.*, Llama-2-Chat-70B, Prometheus, and GPT-3.5-Turbo) achieve low scoring correlation with strong evaluators (*e.g.*, Humans, GPT-4, and Claude-3-Opus). On the other hand, scores provided by strong evaluators highly correlate with each other.

between these two evaluation formats, showing how the scoring decision might change based on a specific evaluation criterion.

2.2.1 Direct Assessment

Direct assessment is the task of mapping an instruction i and response r into a scalar value score s , such as $f_{\text{direct}} : (i, r) \mapsto s$ where $s \in \mathbb{R}$. For the scoring range, we use an integer between 1 and 5.

Prior works have identified several recipes to align the scores provided by evaluator LMs (s_{LM}) and the scores assigned by humans (s_{human}). For instance, Liu et al. [2023a] and Zheng et al. [2023] have shown that it is crucial to add a reference answer a as input to the evaluator LM to maximize the correlation between s_{LM} and s_{human} . Also, Zheng et al. [2023] and Ye et al. [2023] showed that prompting the language model to write verbal feedback v_r before s also improves the correlation between s_{LM} and s_{human} .

Lastly, Ye et al. [2023] and Kim et al. [2023b] showed that by explicitly integrating evaluation criteria e , users can define the standards for model assessment, ensuring evaluations are flexible to specific needs rather than generic qualities. Specifically, e is represented as a score rubric including a description for the criterion itself and a set of descriptions for each score between the scoring range. This is expressed as:

$$f_{\text{direct}} : (i, r, a, e) \mapsto (v_r, s) \quad (2.1)$$

where $s \in \{1, 2, 3, 4, 5\}$

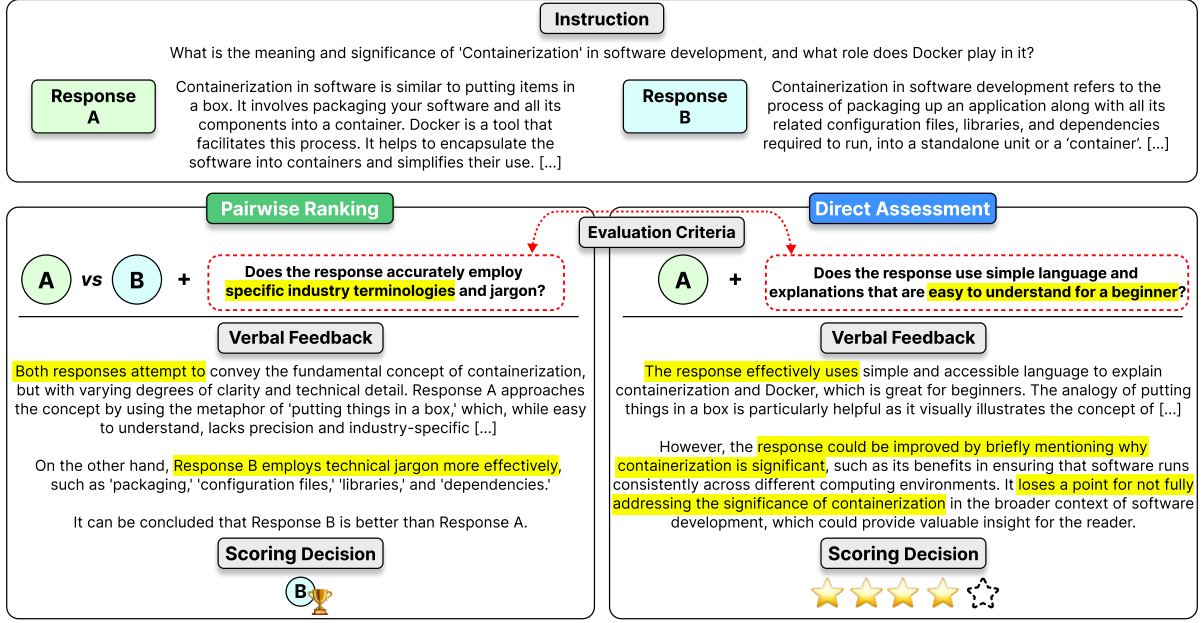


Figure 2.2: Comparison of direct assessment and pairwise ranking. Both responses could be considered decent under the umbrella of ‘helpfulness’. However, the scoring decision might change based on a specific evaluation criterion.

2.2.2 Pairwise Ranking

Pairwise ranking is the task of mapping an instruction i and two pair of responses (r_m, r_n) into either m or n , such as $f_{pair} : (i, r_m, r_n) \mapsto s$ where $s \in \{m, n\}$.

Similar to direct assessment, prior works have identified that integrating a reference answer a and verbal feedback v_{r_m, r_n} into the evaluation pipeline is crucial [Zheng et al., 2023, Li et al., 2023b,a]. In addition, to support granular assessment under custom criterion, we add the evaluation criteria e as input to the evaluator LM [Ye et al., 2023, Kim et al., 2023b]. To the best of our knowledge, we are the first to study such fine-grained evaluation in pairwise ranking settings. This is expressed as:

$$f_{pair} : (i, r_m, r_n, a, e) \mapsto (v_{r_m, r_n}, s) \quad (2.2)$$

where $s \in \{m, n\}$

In pairwise ranking, the evaluation criterion e does not include a set of descriptions for each score; instead, only the description of the evaluation criterion itself. It is noteworthy that the verbal feedback v_{r_m, r_n} compares the commonalities and differences between r_m and r_n concerning e , providing a detailed rationale for the preference decision.

2.3 Training Data

To train a unified evaluator that can perform both direct assessment and pairwise ranking with custom evaluation criteria, we require two complementary training datasets. We leverage the FEEDBACK COLLECTION [Kim et al., 2023b] for direct assessment training and construct a new PREFERENCE COLLECTION for pairwise ranking training. Table 2.1 presents the statistics of both datasets.

Data	PREFERENCE COLLECTION	FEEDBACK COLLECTION
Evaluation Scheme	Pairwise Ranking	Direct Assessment
# Evaluation Criteria	1,000	1,000
# Instructions	20,000	20,000
# Reference Answer	20,000	20,000
# Instances	200,000	100,000
# Verbal Feedback	200,000	100,000

Table 2.1: Statistics of our training datasets, the FEEDBACK COLLECTION and the PREFERENCE COLLECTION. Note that the 1K evaluation criteria, 20K instructions, and 20K reference answers are *shared* among the two datasets. Both datasets have an equal number of scoring decisions (“A” or “B”; 100K each & 1-5; 20K each) to prevent unintended biases after training.

2.3.1 Feedback Collection

The FEEDBACK COLLECTION [Kim et al., 2023b] is a direct assessment feedback dataset designed to train evaluator LMs that can assess responses based on custom evaluation criteria. This dataset contains 100,000 instances covering 20,000 unique instructions and 1,000 evaluation criteria. Each instance includes:

- An instruction i that specifies the task
- A reference answer a that serves as a gold standard
- A response r to be evaluated
- An evaluation criterion e with detailed score rubric descriptions for scores 1-5
- Verbal feedback v_r explaining the assessment
- A score $s \in \{1, 2, 3, 4, 5\}$ representing the quality of the response

The FEEDBACK COLLECTION provides fine-grained evaluation criteria that go beyond generic qualities like helpfulness and harmlessness, enabling evaluator LMs to assess responses based on specific user-defined standards. For each instruction, the dataset includes five responses, each corresponding to a scoring decision between 1 and 5.

2.3.2 Preference Collection

Popular pairwise ranking datasets such as HH-RLHF [Bai et al., 2022] or Ultra Feedback [Cui et al., 2023] do not include an evaluation criterion e and high-quality verbal feedback v_{r_m, r_n} . To train an evaluator LM that could assess based on such criteria, we construct the PREFERENCE COLLECTION, including 1,000 evaluation criteria.

We apply two modifications to the FEEDBACK COLLECTION to create the PREFERENCE COLLECTION:

Pairing Responses Since the FEEDBACK COLLECTION includes five responses for each instruction, each corresponding to a scoring decision between 1 and 5, we pair two out of the five responses, resulting

in a total of ten combinations per instruction. Using the existing scoring decisions for each response, we determine which response is better and assign a new scoring decision for that pair (*i.e.*, “Response A is better” or “Response B is better”).

Generating Comparative Feedback To generate new verbal feedback v_{r_m, r_n} for each pair of responses, we prompt GPT-4-1106 to identify the commonalities and differences between the two responses. This comparative feedback focuses on how the two responses differ with respect to the evaluation criterion, providing detailed rationale for the preference judgment.

The resulting PREFERENCE COLLECTION contains 200,000 instances with 1,000 evaluation criteria, making it the first pairwise ranking dataset that includes instance-wise evaluation criteria beyond basic qualities such as helpfulness and harmlessness. Importantly, the 1,000 evaluation criteria, 20,000 instructions, and 20,000 reference answers are *shared* between the FEEDBACK COLLECTION and the PREFERENCE COLLECTION, ensuring consistency across the two training datasets.

2.4 Weight Merging Strategy

To create a unified evaluator LM capable of performing both direct assessment and pairwise ranking, we explore different training strategies and find that weight merging provides the most effective solution. We compare four primary approaches:

2.4.1 Training Methods

Prompting Prompting involves querying an LM to make judgments in a specified evaluation format without any training. We employ Llama-2-Chat-7,13,70B [Touvron et al., 2023]; Mistral-7B-Instruct-v0.2 [Jiang et al., 2023a]; and Mixtral-8x7B-Instruct-v0.1 [Jiang et al., 2024] as our baselines. It is worth noting that models not explicitly trained on feedback data often fail to generate responses in the required format, making it extremely difficult to parse scoring decisions.

Single-Format Training Single-format training involves training a base model θ either on a direct assessment feedback dataset D_d or a pairwise ranking feedback dataset D_p . For single-format trained evaluator LMs, we test Prometheus-7,13B [Kim et al., 2023b] (direct assessment); UltraRM-13B [Cui et al., 2023] (pairwise ranking); and PairRM-0.4B [Jiang et al., 2023c] (pairwise ranking). In addition, we also report the performances of single-format training Mistral-7B-Instruct-v0.2 and Mixtral-8x7B-Instruct-v0.1 on either direct assessment or pairwise ranking.

Joint Training Joint training involves training a base model θ on both a direct assessment feedback dataset D_d and a pairwise ranking feedback dataset D_p simultaneously through multi-task learning. This enables the resulting evaluator LM to function across both evaluation formats. For jointly trained evaluator LMs, we test Auto-J [Li et al., 2023a]. In addition, we report the performances of jointly training Mistral-7B and Mixtral-8x7B on both direct assessment and pairwise ranking.

Weight Merging Weight merging involves training two models, θ_d and θ_p , separately on a direct assessment feedback dataset D_d and a pairwise ranking feedback dataset D_p . Then, the final evaluator LM θ_{final} is obtained by merging θ_d and θ_p . For linear merging, this is expressed as:

Evaluation Method	Benchmark	Metrics	Judgment Source	Reference Answer	# Score Rubrics	# Instructions	# Judgments
Direct Assessment	Vicuna Bench	Correlation	Proprietary LMs	Y	80	80	320
	MT Bench	Correlation	Proprietary LMs	Y	80	80	320
	FLASK	Correlation	Proprietary LMs & Humans	Y	12	200	2,000
	Feedback Bench	Correlation	Proprietary LMs	Y	200	200	1,000
Pairwise Ranking	HHH Align.	Accuracy	Humans	N	4	221	221
	MT Bench Human Judg.	Accuracy	Humans	N	1	80	3,360
	Auto-J Eval	Accuracy	Humans	N	1	58	1,392
	Preference Bench	Accuracy	Proprietary LMs	Y	200	200	2,000

Table 2.2: Statistics of our evaluation benchmarks to assess the evaluation capabilities of evaluator LMs.

$$\theta_{final} = \alpha \times \theta_d + (1 - \alpha) \times \theta_p \quad (2.3)$$

2.4.2 Merging Techniques

In addition to linear merging, we test five additional variants: Task Arithmetic merging [Ilharco et al., 2022], TIES merging [Yadav et al., 2024], DARE-TIES and DARE-Linear merging [Yu et al., 2023], and SLERP merging [Goddard et al., 2024]. Through extensive ablation experiments, we find that DARE-Linear showed the best performance, and hence we use it to train the PROMETHEUS 2 (7B and 8×7B) models.

We choose Mistral-7B [Jiang et al., 2023a] and Mixtral-8x7B [Jiang et al., 2024] as our base models due to their strong instruction-following capabilities and open-source availability. We train separate models on the FEEDBACK COLLECTION and the PREFERENCE COLLECTION, then merge their weights using DARE-Linear merging to obtain our resulting models, PROMETHEUS 2 (7B and 8×7B).

2.5 Experimental Setup

We evaluate the performance of PROMETHEUS 2 models on eight benchmarks covering both direct assessment and pairwise ranking scenarios. Table 2.2 presents the statistics of all evaluation benchmarks.

2.5.1 Direct Assessment Benchmarks

We evaluate on four direct assessment benchmarks:

- **Vicuna Bench** [Chiang et al., 2023]: A single-turn chat benchmark that includes 80 test prompts, 80 hand-crafted score rubrics from Kim et al. [2023b], and 320 responses obtained by WizardLM-13B, Vicuna-13B, Llama-2-Chat-13B, GPT-3.5-Turbo-0613.
- **MT Bench** [Zheng et al., 2023]: A multi-turn chat benchmark that consists of 80 test prompts, 80 hand-crafted score rubrics from Kim et al. [2023b], and 320 responses obtained by WizardLM-13B, Vicuna-13B, Llama-2-Chat-13B, GPT-3.5-Turbo-0613.
- **FLASK** [Ye et al., 2023]: A fine-grained evaluation benchmark comprised of 200 test prompts, 12 score rubrics, and 2,000 responses acquired from Alpaca-7B, Vicuna-13B, Bard, GPT-3.5-Turbo-0613. In addition to scores from proprietary LMs, this benchmark also includes scores marked by human evaluators.
- **Feedback Bench** [Kim et al., 2023b]: The test set of the FEEDBACK COLLECTION with 1,000 score rubrics, 200 instructions, and 1,000 responses that do not overlap with the training data.

In direct assessment, we conduct **reference-based** evaluations by appending the reference answer as input. We use **Pearson**, **Spearman**, and **Kendall-Tau** as performance metrics to measure scoring correlations against reference evaluators (human evaluators and proprietary LMs such as GPT-4-1106 and Claude-3-Opus).

2.5.2 Pairwise Ranking Benchmarks

We evaluate on four pairwise ranking benchmarks:

- **HHH Alignment** [Askell et al., 2021]: A benchmark consisting of 221 prompts; 4 score rubrics (helpfulness, harmlessness, honesty, and other) and 221 response pairs (graded as ‘win’ or ‘lose’) judged by human evaluators.
- **MT Bench Human Judgment** [Zheng et al., 2023]: A benchmark that shares the same 80 prompts as MT-Bench. In addition, it provides 3,360 response pairs (graded as ‘win’, ‘tie’, or ‘lose’) judged by human evaluators.
- **Auto-J Eval** [Li et al., 2023a]: A benchmark consisted of 58 prompts and 1,392 response pairs (graded as ‘win’, ‘tie’, or ‘lose’) judged by human evaluators. This benchmark is used as the in-domain test set of Auto-J.
- **Preference Bench**: Our in-domain test set for the PROMETHEUS models. Similar to how the PREFERENCE COLLECTION was constructed from the FEEDBACK COLLECTION, we adapt the FEEDBACK BENCH by pairing two out of the five responses, resulting in a test set with 200 prompts, 2,000 response pairs (graded as ‘win’ or ‘lose’), and 200 evaluation criteria.

In pairwise ranking, we conduct **reference-free** evaluations. Based on judgments assigned by humans, we use **accuracy** as our metric to measure agreement between evaluator LMs and humans. For benchmarks that include a ‘tie’ option (MT Bench Human Judgment and Auto-J Eval), we evaluate in two ways: by excluding all ‘tie’ options for pairwise ranking (denoted as ‘**w/o tie**’), or by using direct assessment where responses scored as ‘ties’ are grouped, and pairwise rankings are applied to the remaining responses with differing scores (denoted as ‘**w/ tie**’).

2.6 Results

We evaluate PROMETHEUS 2 models against a comprehensive set of baselines including prompted base models, single-format trained evaluators, jointly trained evaluators, and proprietary LMs. The results demonstrate that PROMETHEUS 2 achieves state-of-the-art performance among open-source evaluator LMs on both direct assessment and pairwise ranking tasks.

2.6.1 Direct Assessment Performance

Table 2.3 presents the direct assessment results. The scoring decisions of PROMETHEUS 2 models (7B and 8×7B), GPT-4-1106, Claude-3-Opus, and human evaluators all strongly correlate with each other, yielding Pearson correlations higher than 0.5 regardless of the reference evaluator and benchmark. On the other hand, base LMs, single-format trained LMs, and jointly trained LMs show lower correlations, mostly falling below 0.5.

Evaluator LM	VICUNA BENCH		MT BENCH		FLASK			Feedback Bench
	GPT-4-1106	Claude-3-Opus	GPT-4-1106	Claude-3-Opus	GPT-4-1106	Claude-3-Opus	Humans	GPT-4-0613
LLAMA2-CHAT 7B	0.205	0.243	0.036	0.055	0.317	0.256	0.299	0.523
LLAMA2-CHAT 13B	0.185	0.141	-0.042	-0.002	0.239	0.247	0.263	0.545
LLAMA2-CHAT 70B	0.350	0.463	0.178	0.228	0.388	0.402	0.317	0.592
MISTRAL-INSTRUCT-7B	0.486	0.561	0.284	0.396	0.448	0.437	0.377	0.586
MIXTRAL-INSTRUCT-8x7B	0.566	0.579	0.551	0.539	0.483	0.495	0.420	0.673
PROMETHEUS-7B	0.484	0.528	0.378	0.382	0.352	0.331	0.348	0.847
PROMETHEUS-13B	0.492	0.534	0.404	0.477	0.462	0.470	0.449	0.860
AUTO-J (13B)	0.351	0.262	0.432	0.375	0.430	0.370	0.473	0.637
PROMETHEUS-2-7B	<u>0.666</u>	0.654	<u>0.548</u>	<u>0.517</u>	<u>0.617</u>	<u>0.561</u>	<u>0.545</u>	<u>0.882</u>
PROMETHEUS-2-8x7B	0.685	<u>0.635</u>	0.665	0.614	0.659	0.626	0.555	0.898
GPT-3.5-TURBO-0613	0.335	0.349	0.183	0.194	0.437	0.396	0.450	0.594
GPT-4-1106	/	0.694	/	0.717	/	0.736	0.679	0.753
CLAUDE-3-OPUS	0.694	/	0.717	/	0.736	/	0.573	0.788

Table 2.3: **Direct Assessment Results.** Pearson correlations between reference evaluators (listed on top) and evaluator LMs. The best comparable statistics are **bolded** and second best underlined except proprietary LMs. Note that the Feedback Bench is an in-domain test set of the PROMETHEUS models.

Notably, PROMETHEUS 2 models outperform Prometheus and Auto-J by at least 0.2 units across benchmarks in their correlation with proprietary LMs. Moreover, on the FLASK benchmark, while the correlation between humans and GPT-4 is 0.679, the highest correlation previously achieved by Prometheus-13B with humans was 0.449. PROMETHEUS-2-8x7B achieves a correlation of 0.555 with humans, halving the gap between open-source evaluators and proprietary LMs.

These results demonstrate that PROMETHEUS 2 models can effectively simulate both human judgments and proprietary LM evaluations in direct assessment scenarios. The substantial improvement over previous open-source evaluators highlights the effectiveness of the weight merging approach for creating unified evaluator LMs.

2.6.2 Pairwise Ranking Performance

Table 2.4 presents the pairwise ranking results. On all four benchmarks, the PROMETHEUS 2 models achieve the highest scores among open-source evaluators, showing that they can effectively simulate human judgments in pairwise ranking scenarios.

Notably, while HHH Alignment is an in-domain test set for Pair RM, and Auto-J Eval is for Auto-J, PROMETHEUS-2-8x7B achieves competitive or higher scores on these benchmarks. This shows that training a large LM (*e.g.*, Mixtral-8x7B) with fine-grained feedback data can be an effective strategy to obtain a robust evaluator LM that can generalize beyond its training data. Moreover, the PROMETHEUS 2 models at least halve the performance gap with proprietary LMs compared to existing open-source evaluator LMs on out-of-domain test sets.

The strong performance on both in-domain (Preference Bench) and out-of-domain benchmarks demonstrates the generalization capability of PROMETHEUS 2 models in pairwise ranking scenarios.

2.6.3 Comparison with Proprietary LMs

When comparing PROMETHEUS 2 models with proprietary LMs, we observe that:

Evaluator LM	HHH ALIGNMENT					MT BENCH HUMAN JUDG.		AUTO-J EVAL		Preference Bench
	Help.	Harm.	Hon.	Other	Total Avg.	w/ TIE	w/o TIE	w/ TIE	w/o TIE	Instance-wise Criteria
LLAMA2-CHAT 7B	55.93	62.07	49.18	62.79	57.01	46.68	50.39	45.76	45.73	58.60
LLAMA2-CHAT 13B	71.19	77.59	60.66	62.79	68.33	51.22	49.61	47.84	43.28	63.00
LLAMA2-CHAT 70B	62.71	81.03	65.57	65.12	68.78	55.14	60.88	53.38	50.64	64.70
MISTRAL-INSTRUCT-7B	59.32	68.97	63.93	81.40	67.42	53.81	63.82	53.88	60.94	79.40
MIXTRAL-INSTRUCT-8X7B	83.05	<u>87.93</u>	67.21	69.77	77.38	51.85	<u>71.42</u>	53.81	73.50	84.00
PAIR RM (0.4B)	<u>84.75</u>	84.48	<u>80.33</u>	90.70	<u>84.62</u>	-	59.00	-	59.05	81.80
ULTRA RM (13B)	86.44	79.31	81.97	<u>88.37</u>	83.71	-	56.00	-	59.85	86.97
AUTO-J (13B)	77.97	79.31	70.49	74.42	75.57	42.56	69.12	43.46	<u>76.64</u>	81.35
PROMETHEUS-2-7B	72.78	79.31	77.05	76.74	74.66	50.45	70.78	54.96	75.07	93.25
PROMETHEUS-2-8X7B	<u>84.75</u>	96.55	81.97	76.74	85.52	<u>55.07</u>	71.96	58.41	79.98	<u>90.65</u>
GPT-3.5-TURBO-0613	77.97	81.03	77.05	67.44	76.47	54.65	69.41	45.98	72.13	75.05
GPT-4-1106-PREVIEW	89.83	96.55	91.80	83.72	90.95	60.38	79.90	52.80	83.12	85.50
CLAUDE-3-OPUS	91.53	100.00	91.80	95.35	94.57	55.35	77.65	60.70	82.92	89.85

Table 2.4: **Pairwise Ranking Results.** Accuracy on human preference datasets. The best comparable accuracies are **bolded** and second best underlined except proprietary LMs. Note that HHH Alignment is an in-domain test set for PairRM, Auto-J Eval is an in-domain test set for Auto-J, and the Preference Bench is an in-domain test set for Prometheus-2 models.

- On direct assessment benchmarks, PROMETHEUS-2-8X7B achieves Pearson correlations with human evaluators (0.555 on FLASK) that approach those of GPT-4-1106 (0.679), substantially closing the gap that existed with previous open-source evaluators.
- On pairwise ranking benchmarks, PROMETHEUS-2-8X7B achieves accuracy scores that are competitive with GPT-4-1106 on several benchmarks, particularly on MT Bench Human Judgment and Auto-J Eval.
- The correlation between PROMETHEUS 2 models and proprietary LMs (GPT-4, Claude-3) is consistently high (often > 0.6), indicating that these open-source evaluators can effectively simulate the behavior of proprietary evaluators.

These results demonstrate that PROMETHEUS 2 successfully bridges the gap between open-source and proprietary evaluator LMs, providing a transparent, controllable, and affordable alternative while maintaining competitive evaluation performance.

2.7 Analysis

To understand the effectiveness of our proposed weight merging method and to gain insights into how different training strategies affect evaluator LM performance, we conduct comprehensive analyses addressing three key research questions.

2.7.1 Ablation Studies

Weight Merging vs Joint Training

Table 2.5 compares the performance of evaluator LMs trained via weight merging and joint training, alongside prompting and single-format training baselines.

The results reveal several important findings:

Training Method	DIRECT ASSESSMENT BENCHMARKS				PAIRWISE RANKING BENCHMARKS				
	Vicuna Ben.	MT Ben.	FLASK	Average	HHH Align.	MT Ben. H.J.	Auto-J Eval	Average	
<i>Mistral-Instruct-7B</i>									
PROMPTING	0.486	0.284	0.480	0.417	67.42	63.82	60.94	64.06	
DIRECT ASSESSMENT ONLY	0.537	0.561	<u>0.519</u>	<u>0.539</u>	73.33	56.76	64.38	64.82	
PAIRWISE RANKING ONLY	-	-	-	-	<u>78.73</u>	<u>67.06</u>	72.03	72.61	
JOINT TRAINING	<u>0.548</u>	0.450	0.457	0.485	80.09	65.49	<u>73.60</u>	<u>73.06</u>	
WEIGHT MERGING	0.666	<u>0.548</u>	0.659	0.624	74.66	70.78	75.07	73.50	
<i>Mixtral-Instruct-8x7B</i>									
PROMPTING	0.566	0.551	0.507	0.541	77.38	<u>71.42</u>	73.55	74.56	
DIRECT ASSESSMENT ONLY	0.625	<u>0.664</u>	0.587	<u>0.625</u>	74.21	53.14	65.85	64.40	
PAIRWISE RANKING ONLY	-	-	-	-	<u>84.16</u>	66.27	<u>75.66</u>	<u>75.36</u>	
JOINT TRAINING	<u>0.628</u>	0.560	<u>0.596</u>	0.595	82.35	68.73	74.78	75.29	
WEIGHT MERGING	0.685	0.665	0.659	0.670	85.52	71.96	79.98	79.15	

Table 2.5: **Single-Format Training vs Joint Training vs Weight Merging.** Pearson correlations between evaluator LMs trained with different methods and GPT-4-1106. Evaluator LMs trained with weight merging outperform single-format-trained and jointly-trained evaluator LMs across multiple benchmarks.

Negative Transfer in Joint Training Surprisingly, evaluator LMs trained via joint training often show lower performance compared to those trained only in single-format, which indicates *negative task transfer*. Specifically, evaluator LMs trained only on direct assessment formats obtain higher correlations compared to their jointly trained counterparts across different model scales. Similarly, evaluator LMs trained solely on pairwise ranking formats achieve higher average accuracy compared to those trained on multiple tasks, particularly when using Mixtral-8x7B as the base model.

Positive Transfer in Weight Merging In contrast, evaluator LMs trained via weight merging show superior performance not only compared to jointly trained evaluator LMs but also single-format trained evaluator LMs, indicating *positive task transfer*. The weight-merged models achieve the best performance on both direct assessment and pairwise ranking benchmarks across both base model sizes (Mistral-7B and Mixtral-8x7B).

These findings suggest that weight merging successfully addresses the negative transfer problem inherent in joint training by allowing each model to specialize in its respective evaluation format before combining their knowledge through parameter merging.

Unifying Formats vs Ensembling

While we empirically find that weight merging is effective, the underlying reason requires further investigation. A natural hypothesis is that this effectiveness results from the ensembling effect of combining multiple models. To test this hypothesis, we conduct an ablation experiment where we train multiple evaluator LMs with different random seeds and merge them.

Specifically, we merge two evaluator LMs trained on direct assessment formats (denoted as ‘Direct Assessment & Direct Assessment’) and two evaluator LMs trained on pairwise ranking formats (denoted as ‘Pairwise Ranking & Pairwise Ranking’). We use Mistral-7B-Instruct as our base model. The results are presented in Table 2.6.

The results demonstrate that across multiple benchmarks, merging evaluator LMs trained on the

Training Data Evaluation Format	DIRECT ASSESSMENT BENCHMARKS				PAIRWISE RANKING BENCHMARKS			
	Vicuna Ben.	MT Ben.	FLASK	Average	HHH Align.	MT Ben. H.J.	Auto-J Eval	Average
NO TRAINING (PROMPTING)	0.486	0.284	0.480	0.417	67.42	63.82	60.94	64.06
DIRECT ASSESSMENT ONLY	0.537	0.561	<u>0.519</u>	<u>0.539</u>	73.33	56.76	64.38	64.82
PAIRWISE RANKING ONLY	-	-	-	-	78.73	<u>67.06</u>	72.03	<u>72.61</u>
DIRECT ASSESSMENT & DIRECT ASSESSMENT	<u>0.552</u>	0.493	0.505	0.517	73.30	55.00	63.69	64.13
PAIRWISE RANKING & PAIRWISE RANKING	-	-	-	-	<u>78.70</u>	65.20	<u>72.72</u>	72.21
DIRECT ASSESSMENT & PAIRWISE RANKING	0.666	<u>0.548</u>	0.659	0.624	74.66	70.78	75.07	73.50

Table 2.6: **Unifying Formats vs Ensembling.** Pearson correlations with GPT-4-1106 (Vicuna Bench, MT Bench, FLASK) and agreement with human evaluators (HHH Alignment, MT Bench Human Judgment, Auto-J Eval). Merging models trained with the same evaluation formats (ensembling) underperforms merging models trained with different formats (unifying formats).

same evaluation format does not enhance evaluation performance. Specifically:

- Merging two evaluator LMs trained on direct assessment (Direct Assessment & Direct Assessment) achieves an average Pearson correlation of 0.517 on direct assessment benchmarks, which is *lower* than the single-format baseline (0.539).
- Merging two evaluator LMs trained on pairwise ranking (Pairwise Ranking & Pairwise Ranking) achieves an average accuracy of 72.21% on pairwise ranking benchmarks, which is also *lower* than the single-format baseline (72.61%).

In contrast, merging two evaluator LMs trained on different formats (Direct Assessment & Pairwise Ranking) results in superior performance compared to all other settings, achieving 0.624 average Pearson correlation on direct assessment and 73.50% average accuracy on pairwise ranking.

This indicates that the beneficial task transfer in weight merging arises from integrating different evaluation formats, not from ensembling multiple models. The complementary nature of direct assessment and pairwise ranking knowledge appears to be crucial for the effectiveness of the weight merging approach.

2.7.2 Optimal Merging Weight

To explore how training on direct assessment feedback data influences pairwise ranking accuracy and vice versa, we experiment by adjusting the α value during linear merging. We evaluate the average performance using all eight benchmarks in our experiments. To illustrate the average performance, we adjust the scale by multiplying the Pearson correlations from direct assessment, which originally range from 0 to 1, by 100 before averaging them with the pairwise ranking accuracy.

The results are shown in Figure 2.3. Several key observations emerge:

Optimal Weight for Direct Assessment For direct assessment benchmarks, evaluator LMs obtain optimal performance when α is set to 0.5. This indirectly indicates that both pairwise ranking and direct assessment feedback data contribute equally to direct assessment performance.

Optimal Weight for Pairwise Ranking On the other hand, for pairwise ranking benchmarks, the performance is optimal when α is set to 0.3. This implies that while both benefit each other, training on pairwise ranking improves direct assessment performance more significantly than the reverse.

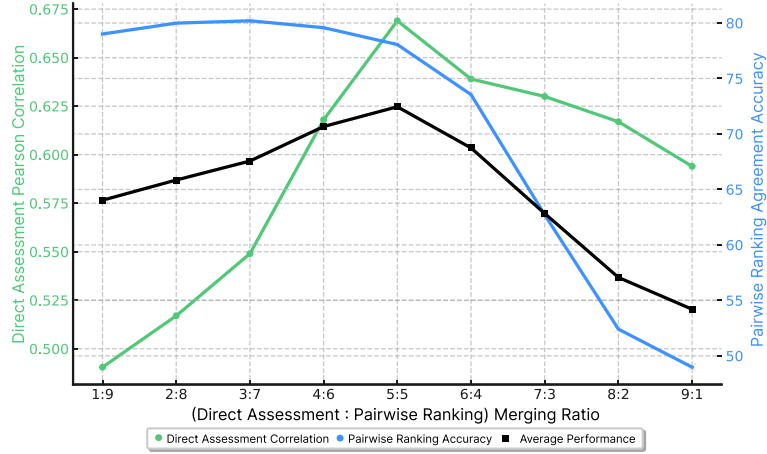


Figure 2.3: **When merging models, the influence of relative evaluation on absolute evaluation is greater than the influence of absolute evaluation on relative evaluation.** Performance of Direct Assessment (colored in green) and Pairwise Ranking (colored in blue) when altering the α value to merge evaluator LMs trained on different formats.

Asymmetric Positive Transfer The asymmetry in optimal α values suggests that pairwise ranking knowledge transfers more effectively to direct assessment than vice versa. This finding aligns with the intuition that pairwise ranking requires understanding fine-grained differences between responses, which may be more informative for developing robust evaluation capabilities.

Robust Performance Across α Values The average performance (shown in black) remains relatively stable across a wide range of α values (0.2 to 0.7), indicating that weight merging is robust to the choice of merging weight within a reasonable range.

These findings provide valuable insights for practitioners seeking to train unified evaluator LMs: while both evaluation formats contribute to overall performance, the optimal merging strategy should slightly favor pairwise ranking weights to maximize performance across both evaluation formats.

Chapter 3. Continual Training for Benchmark Adaptation

In this chapter, we explore how Prometheus-2 can be adapted to specific benchmarks through continual training, demonstrating its flexibility and practical applicability for affordable internal evaluation pipelines. We investigate two complementary strategies to improve open-source evaluator performance: self-consistency decoding and continual feedback training, and validate these approaches using BiGGen-Bench as a test benchmark.

3.1 Motivation: Affordable Internal Evaluation Pipelines

For transparent and accessible evaluations, it is crucial to develop strategies for employing open-source evaluator LMs [Kim et al., 2023b, Lee et al., 2024, Kim et al., 2024]. While Prometheus-2 demonstrates strong correlation with human judgments on standard benchmarks, proprietary LMs like GPT-4 still maintain an edge in evaluation performance. This raises an important practical question for LM developers.

In practice, LM developers maintain a fixed test set to monitor the performance of the models they are developing. As they refine their models, they may adjust configurations such as model size, learning rate, or training objective, and then evaluate performance differences to determine the optimal settings. When employing generation benchmarks, using GPT-4 as an evaluator incurs a constant cost proportional to the number of test runs, which becomes unaffordable as the number of runs increases. This raises a natural question: whether accumulated feedback on a single benchmark can be leveraged to train an open-source evaluator LM that performs well on that benchmark, thereby establishing an affordable internal evaluation pipeline.

To answer this question, we investigate two complementary strategies: **self-consistency decoding** [Wang et al., 2022] and **continual training** [Scialom et al., 2022]. These approaches aim to narrow the gap between open-source and proprietary evaluator LMs, enabling more affordable and transparent evaluation workflows.

3.2 BiGGen-Bench: A Validation Benchmark

3.2.1 Brief Overview

To validate our continual training methodology, we use BiGGen-Bench as a test benchmark. BiGGen-Bench is a comprehensive benchmark designed to evaluate the instruction-following capabilities of large language models. The benchmark consists of 77 diverse tasks across 9 core capabilities (e.g., reasoning, multilingual understanding, coding), with 765 task instances in total. Each instance includes human-defined evaluation rubrics that specify detailed criteria for assessment.

For our experiments, we leverage the evaluation results from BiGGen-Bench, which assessed 103 different language models using proprietary evaluators (GPT-4-1106, GPT-4-Turbo, Claude-3-Opus). These evaluations provide a rich dataset of feedback that we can use to continually train Prometheus-2.

3.2.2 Why BiGGen-Bench for This Study

BiGGen-Bench is particularly well-suited for validating our continual training methodology for several reasons. First, it provides a large-scale collection of GPT-4 judgments (78,795 evaluations across 103 models), enabling us to test whether accumulated feedback can effectively train an open-source evaluator. Second, the benchmark includes human annotations for a subset of models, allowing us to measure correlation with human judgments. Third, the diversity of tasks and capabilities in BiGGen-Bench provides a comprehensive testbed for examining whether continual training improves evaluation performance broadly or only on specific task types.

3.3 Continual Training Methodology

3.3.1 Training Data Setup

To test our hypothesis about continual training, we divide the 78,795 judgments made by GPT-4-1106 on BiGGen-Bench into two groups: a training set of 50,490 judgments (from 66 response LMs) and a test set of 28,305 judgments (from the remaining 37 response LMs). We use the training set to continually train Prometheus-2 and measure evaluation performance on the test set.

This experimental setup creates a realistic scenario: the continually trained model (which we call Prometheus-2-BGB) has encountered the 765 task inputs during training (*i.e.*, seen inputs), but has not seen the responses from the 37 test LMs it will evaluate (*i.e.*, unseen responses). Among these 37 test LMs, four have human annotations (Llama-2-13b-hf, Mistral-7B-Instruct-v0.2, Mixtral-8x7B-Instruct-v0.1, gpt-3.5-turbo-0125), enabling us to measure correlation with human judgments.

3.3.2 Prometheus-2-BGB Training

We continually train the Prometheus-2-8x7B model on the training set using supervised fine-tuning with LoRA (Low-Rank Adaptation) [Hu et al., 2021]. The training data consists of GPT-4-1106’s feedback on the 66 training LMs across all 765 BiGGen-Bench instances. Each training example follows Prometheus-2’s standard format, with an instruction, response, evaluation rubric, and reference answer, along with GPT-4-1106’s score and feedback.

The continual training process allows Prometheus-2-BGB to learn the evaluation patterns and preferences exhibited by GPT-4-1106 on this specific benchmark, while building upon the general evaluation capabilities acquired during Prometheus-2’s original training.

3.3.3 Hyperparameters

Table 3.1 lists the hyperparameters used to train Prometheus-2-BGB. We use LoRA for parameter-efficient fine-tuning with a rank of 256 and alpha of 512, targeting the query, key, value, output projection layers and the language model head. Training is conducted for 1 epoch with a learning rate of 1e-5 and batch size of 8. We set the maximum sequence length to 4096 tokens and use bfloat16 precision. All random seeds are set to 42 for reproducibility.

The 66 training models span diverse architectures, sizes (from 0.5B to 72B parameters), and model families (including Llama, Mistral, Mixtral, Qwen, Yi, Gemma, and others), providing broad coverage of the LM landscape.

Table 3.1: Hyperparameters used to train Prometheus-2-BGB-8x7B.

Base Model	prometheus-eval/prometheus-8x7b-v2.0
Torch dtype	bfloat16
Epoch	1
Train Data	BiGGen-Bench Results
Max Seq Length	4096
Learning Rate	1e-5
Train Batch Size	8
PEFT	True
Lora_r	256
Lora_alpha	512
Lora_Dropout	0.1
Lora Target Module	Q proj, K proj, V proj, O proj, W proj, LM.Head
Random Seed	42
Training Method	Supervised Fine-tuning

3.4 Self-Consistency Decoding

3.4.1 Motivation

Self-consistency decoding was originally proposed by Wang et al. [2022] to enhance the problem-solving abilities of language models. The approach involves sampling multiple generations and taking a majority vote to decide the final prediction. We adapt this technique to improve the evaluation capabilities of Prometheus-2.

The intuition behind applying self-consistency to evaluation is that evaluator LMs, while deterministic in their training objective, can exhibit variance in their assessments due to the stochastic nature of sampling. By generating multiple independent evaluations and aggregating them, we can potentially achieve more robust and reliable assessments.

3.4.2 Implementation

For our experiments, we sample N responses from the evaluator LM (with $N \in \{1, 3, 5\}$) and compute the average score as the final evaluation. This differs slightly from the original self-consistency approach, which uses majority voting for classification tasks. Since our evaluation task produces continuous scores (typically on a 1-5 scale), averaging provides a natural aggregation method.

3.4.3 Results

As shown in Table 3.2, increasing the number of samples from 1 to 3 enhances Prometheus-2’s correlation with human evaluators ($0.471 \rightarrow 0.502$), indicating an improvement in evaluation precision. However, further increasing the number of samples from 3 to 5 results in minimal additional improvements ($0.502 \rightarrow 0.503$).

We conjecture that expanding the number of samples from 1 to 3 allows Prometheus-2 to benefit from the diversity of the generated feedback, but increasing beyond this point is less effective. Considering that evaluator LMs must maintain consistent gradings even though they are stochastic in nature, sampling three responses appears to be the “sweet spot” for balancing diversity and consistency.

Table 3.2: Correlation between evaluator LMs and human judgments on BiGGen-Bench. Continual feedback training significantly improves correlation with humans, and self-consistency decoding provides additional gains.

Evaluator LM	N=1	N=3
Prometheus-2-8x7B	0.471	0.502
Prometheus-2-BGB-8x7B	0.577	0.607
Claude-3-Opus	0.578	-
GPT-4-1106	0.597	-
GPT-4-Turbo-2024-04-09	0.623	-

Table 3.3: Correlation between scores from open-source evaluator LMs and proprietary LMs. Continual feedback training enables open-source evaluator LMs to more closely mimic the judgments of proprietary LMs.

Evaluator LM	Prometheus-2	Prometheus-2-BGB	
	N=1	N=1	N=5
Claude-3-Opus	0.688	0.735	0.752
GPT-4-1106	0.688	0.836	0.865
GPT-4-Turbo-2024-04-09	0.704	0.833	0.861

Importantly, self-consistency decoding provides complementary benefits to continual training. When combined with Prometheus-2-BGB, self-consistency further boosts performance, as detailed in the following sections.

3.5 Results

3.5.1 Improved Human Correlation

Table 3.2 presents the correlation between evaluator LMs and human judgments on the four models in BiGGen-Bench with human annotations. Prometheus-2-BGB achieves significantly higher human correlations on average compared to its base model, Prometheus-2 ($0.471 \rightarrow 0.577$, a 22% improvement), and performs on par with Claude-3-Opus (0.578) and close to GPT-4-1106 (0.597) and GPT-4-Turbo-2024-04-09 (0.623).

Moreover, when employing self-consistency decoding (N=3) in conjunction with continual training, Prometheus-2-BGB attains a Pearson correlation of 0.607, which surpasses both Claude-3-Opus and GPT-4-1106. This indicates that self-consistency decoding and continual feedback training provide complementary benefits to enhance the performance of evaluator LMs.

These results demonstrate that continual feedback training can effectively enhance an open-source evaluator’s ability to simulate human judgments on unseen responses, even when those responses come from models not included in the training set.

Table 3.4: Pearson correlations between reference evaluators and evaluator LMs on four standard benchmarks. The best comparable statistics are **bolded** and second best underlined except proprietary LMs. Prometheus-2-BGB maintains or improves performance across all benchmarks.

Evaluator LM	Vicuna Bench		MT Bench		FLASK			Feedback Bench
	GPT-4-1106	Claude-3-Opus	GPT-4-1106	Claude-3-Opus	GPT-4-1106	Claude-3-Opus	Humans	GPT-4-0613
Mistral-Instruct-7B	0.486	0.561	0.284	0.396	0.448	0.437	0.377	0.586
Mixtral-Instruct-8x7B	0.566	0.579	0.551	0.539	0.483	0.495	0.420	0.673
Prometheus-2-7B	0.642	0.610	0.543	0.554	0.645	0.578	0.544	0.878
Prometheus-2-8x7B	<u>0.685</u>	0.635	<u>0.665</u>	<u>0.614</u>	<u>0.659</u>	<u>0.626</u>	<u>0.555</u>	0.898
Prometheus-2-BGB-8x7B (Ours)	0.777	<u>0.618</u>	0.773	0.619	0.764	0.635	0.649	<u>0.890</u>
GPT-3.5-Turbo-0613	0.335	0.349	0.183	0.194	0.437	0.396	0.450	0.594
GPT-4-1106	/	0.694	/	0.717	/	0.736	0.679	0.753
Claude-3-Opus	0.694	/	0.717	/	0.736	/	0.573	0.788

3.5.2 Improved GPT-4 Mimicry

Beyond simulating human judgments, an important practical consideration is whether continual training enables open-source evaluators to mimic proprietary LMs effectively. Table 3.3 shows the correlation between Prometheus-2-BGB and three proprietary evaluators (Claude-3-Opus, GPT-4-1106, GPT-4-Turbo-2024-04-09) on the 37 test models.

The results show significant improvements across all proprietary LMs. The correlation with Claude-3-Opus improves from 0.688 to 0.735, with GPT-4-1106 from 0.688 to 0.836 (a 21% increase), and with GPT-4-Turbo from 0.704 to 0.833. These substantial gains demonstrate that continual training on GPT-4 feedback enables Prometheus-2-BGB to effectively learn and replicate the evaluation patterns of proprietary models.

Moreover, when using self-consistency decoding (N=5) in conjunction with continual training, the correlation between Prometheus-2-BGB and GPT-4-1106 reaches 0.865. This exceptionally high correlation indicates that Prometheus-2-BGB can closely mimic GPT-4-1106’s assessments on responses it was not exposed to during training, making it a viable alternative for affordable internal evaluation pipelines.

3.5.3 No Catastrophic Forgetting

In continual learning settings, it is crucial to verify that the model does not suffer from catastrophic forgetting [McCloskey and Cohen, 1989, Kirkpatrick et al., 2017, Jang et al., 2021]—the phenomenon where performance on previously learned tasks degrades after training on new tasks. Since Prometheus-2-BGB was trained on feedback from GPT-4-1106 on BiGGen-Bench, we must check whether its evaluation performance on other benchmarks remains intact.

To measure this, we employ four benchmarks following the evaluation protocol of Prometheus-2 [Kim et al., 2024]: Vicuna Bench [Chiang et al., 2023], MT Bench [Zheng et al., 2023], FLASK [Ye et al., 2023], and Feedback Bench [Kim et al., 2023b]. The first three are benchmarks that Prometheus-2 was not trained on (out-of-domain), while Feedback Bench is an in-domain test set for Prometheus-2.

Table 3.4 presents the results. Remarkably, across all benchmarks and reference evaluators, Prometheus-2-BGB’s performance significantly improves compared to its base model, Prometheus-2-8x7B. On Vicuna Bench, correlations improve from 0.685/0.635 to 0.777/0.618 (GPT-4/Claude). On MT Bench, correlations improve from 0.665/0.614 to 0.773/0.619. On FLASK, correlations improve from 0.659/0.626/0.555 to 0.764/0.635/0.649 (GPT-4/Claude/Humans).

Notably, on all benchmarks, Prometheus-2-BGB correlates more closely with GPT-4-1106 compared

Table 3.5: Ranking correlation between widely used benchmarks and BiGGen-Bench when evaluated with different evaluator LMs. The value N denotes the number of overlapping LMs between each benchmark and BiGGen-Bench.

Benchmark	BiGGen Bench		
	Prometheus-2	Prometheus-2	GPT-4-Turbo
	-8x7B	-BGB-8x7B	2024-04-09
MT Bench (N=18)	0.625	0.859	0.830
MMLU (N=29)	0.871	0.910	0.915
LMSys Arena (N=42)	0.879	0.907	0.909

to even Claude-3-Opus. This suggests that the feedback acquired from BiGGen-Bench triggered positive task transfer during continual training, improving assessment performance on other benchmarks as well.

On the in-domain Feedback Bench, the performance degradation of Prometheus-2-BGB compared to Prometheus-2-8x7B is minimal (0.898 to 0.890, less than 1% decrease). This demonstrates that continual training on BiGGen-Bench does not harm the model’s original capabilities, supporting that Prometheus-2-BGB can function as a reliable evaluator across diverse benchmarks.

3.5.4 Improved Ranking Correlation

Beyond point-wise correlation on individual evaluations, we examine whether Prometheus-2-BGB produces reliable rankings of language models. To validate this, we measure the ranking correlation between BiGGen-Bench scores and three representative benchmarks: MT Bench [Zheng et al., 2023], MMLU [Hendrycks et al., 2020], and LMSys Arena [Chiang et al., 2024].

Table 3.5 presents the ranking correlations. Compared to Prometheus-2, Prometheus-2-BGB’s ranking correlations improve substantially across all three benchmarks. On MT Bench, correlation increases from 0.625 to 0.859. On MMLU, it improves from 0.871 to 0.910. On LMSys Arena, it increases from 0.879 to 0.907.

Notably, Prometheus-2-BGB’s ranking correlations are as high as those of GPT-4-Turbo-2024-04-09 (0.830, 0.915, 0.909 respectively), indicating that it can reliably function as a robust evaluator LM for ranking purposes. The particularly high correlation with LMSys Arena (0.907) is encouraging, as LMSys Arena is one of the most widely referenced online LM evaluation benchmarks based on real-user feedback. This suggests that BiGGen-Bench scores from Prometheus-2-BGB could effectively simulate real-world human preferences without requiring time-consuming human annotations [Saunders et al., 2022, Kim et al., 2023a].

3.6 Analysis

3.6.1 Positive Task Transfer

One of the most surprising findings from our continual training experiments is the positive task transfer observed across different benchmarks. Rather than simply memorizing BiGGen-Bench-specific patterns, Prometheus-2-BGB demonstrates improved evaluation capabilities on benchmarks it was never trained on.

As shown in Table 3.4, Prometheus-2-BGB outperforms the base Prometheus-2-8x7B model on Vicuna Bench, MT Bench, and FLASK—all out-of-domain benchmarks. This improvement cannot be attributed to overfitting on BiGGen-Bench, as these benchmarks have different task distributions, evaluation criteria, and response characteristics.

We hypothesize that continual training on the diverse set of 77 tasks in BiGGen-Bench, spanning multiple capabilities (reasoning, multilingual understanding, coding, etc.), helps the model learn more generalizable evaluation patterns. The feedback from GPT-4-1106 across these diverse tasks may have refined Prometheus-2’s understanding of quality assessment in ways that transfer positively to other evaluation scenarios.

This finding has important practical implications: organizations can accumulate feedback on their internal benchmarks and use it to train specialized evaluators that not only excel on those specific benchmarks but also maintain or improve performance on other evaluation tasks.

3.6.2 Self-Consistency Benefits

Our experiments demonstrate that self-consistency decoding provides consistent improvements across different models and evaluation scenarios. For the base Prometheus-2 model, self-consistency (N=3) improves human correlation from 0.471 to 0.502. For Prometheus-2-BGB, it further boosts correlation from 0.577 to 0.607.

The benefits of self-consistency are particularly pronounced when mimicking proprietary LMs. With N=5 samples, Prometheus-2-BGB achieves 0.865 correlation with GPT-4-1106, compared to 0.836 with N=1. This suggests that aggregating multiple evaluations helps smooth out stochastic variance and produces more stable, reliable assessments.

However, we observe diminishing returns beyond N=3 samples when measuring human correlation. This suggests that the optimal number of samples may depend on the specific evaluation task and reference evaluator. For practical applications, N=3 appears to offer a good balance between improved performance and computational cost, as it requires only 3x the inference time compared to single-sample evaluation.

3.6.3 Practical Implications

The continual training methodology demonstrated in this chapter has significant practical implications for LM developers and researchers. Our results show that accumulating GPT-4 feedback on a fixed test set and using it to continually train an open-source evaluator can create an effective internal evaluation pipeline.

The cost savings are substantial. GPT-4 API costs approximately \$0.01 per 1K input tokens and \$0.03 per 1K output tokens. For a benchmark with 765 instances like BiGGen-Bench, a single evaluation run with detailed feedback can cost hundreds of dollars. Running multiple evaluation iterations during model development quickly becomes prohibitively expensive.

In contrast, after the initial investment of collecting GPT-4 feedback (which can be amortized across multiple model development cycles), Prometheus-2-BGB can be run locally at minimal marginal cost. Our experiments show that Prometheus-2-BGB achieves 0.836 correlation with GPT-4-1106 (0.865 with self-consistency), making it a viable replacement for most internal evaluation purposes.

Moreover, the lack of catastrophic forgetting and positive task transfer suggest that continual training is a safe and beneficial approach. Organizations can continually update their evaluators as they

accumulate more feedback, without worrying about degrading performance on existing evaluation tasks.

Finally, the combination of continual training and self-consistency decoding provides a clear path forward for improving open-source evaluators. While individual techniques offer meaningful improvements, their combination yields the best results, narrowing the gap with proprietary evaluators while maintaining transparency and affordability.

Chapter 4. Conclusion and Future Work

This thesis has explored two interconnected contributions to the field of large language model evaluation: the development of Prometheus 2 as a unified open-source evaluator, and the application of continual training methodology to adapt evaluators for specific benchmarks. Together, these works demonstrate that accessible, transparent, and high-quality evaluation is achievable without reliance on proprietary models.

4.1 Summary of Contributions

This work makes two primary contributions that advance the state of open-source LLM evaluation.

4.1.1 Prometheus 2: A Unified Open-Source Evaluator

We introduced Prometheus 2, the first open-source evaluator capable of performing both direct assessment and pairwise ranking in a single unified model. Through innovative application of weight merging techniques, Prometheus 2 combines the strengths of specialized evaluators while maintaining computational efficiency. Key achievements include:

- **Unified evaluation capability:** A single model (7B or 8×7B parameters) that performs both direct assessment (absolute scoring) and pairwise ranking (relative comparison), eliminating the need for multiple specialized models.
- **Strong correlation with human judgments:** Prometheus 2 achieves correlation scores exceeding 0.6 with both human evaluators and GPT-4 across diverse benchmarks, demonstrating reliability comparable to proprietary alternatives.
- **Open-source accessibility:** By releasing model weights, training data (Feedback Collection), and evaluation code publicly, we enable reproducible research and democratize access to high-quality LLM evaluation tools.
- **Practical applicability:** Prometheus 2 has been successfully deployed in various research contexts, proving its utility for reward modeling, response filtering, and automated evaluation pipelines.

This contribution addresses critical challenges in the LLM evaluation ecosystem: the lack of transparency in proprietary evaluators, the high costs associated with API-based evaluation, and the difficulty of reproducing evaluation results across different research contexts.

4.1.2 Continual Training for Benchmark-Specific Adaptation

Building upon Prometheus 2, we developed and validated a continual training methodology that adapts general-purpose evaluators to specific benchmarks. Through the case study of Prometheus-2-BGB (BiGGen-Bench-adapted), we demonstrated:

- **Significant performance improvement:** Continual training on benchmark-specific feedback data yielded a 22% improvement in Pearson correlation with human judgments on BiGGen-Bench, rising from 0.471 to 0.577.

- **Near-GPT-4 performance:** The adapted model achieved performance within 0.03 correlation points of GPT-4-as-a-judge on the target benchmark, while maintaining the cost and accessibility advantages of open-source models.
- **Preservation of general capabilities:** Evaluation on held-out benchmarks confirmed minimal catastrophic forgetting, with the adapted model maintaining strong performance on tasks it was not specifically trained for.
- **Data efficiency:** Meaningful adaptation was achieved with 50,490 training instances (GPT-4 feedback on 66 models across 765 BiGGen-Bench tasks), demonstrating the feasibility of creating benchmark-specific evaluators.
- **Practical framework:** We established a replicable methodology for organizations to develop internal evaluators tailored to their specific evaluation needs, using Prometheus 2 as a strong foundation.

This contribution provides a concrete path for practitioners to leverage open-source evaluators effectively, adapting them to domain-specific requirements while maintaining the benefits of transparency and reproducibility.

Together, these contributions advance three fundamental principles in LLM evaluation: *fairness* through consistent and reproducible judgments, *transparency* through open-source availability, and *accessibility* through reduced computational and financial costs. By demonstrating that open-source evaluators can match or approach the performance of proprietary alternatives, this work challenges the assumption that high-quality evaluation necessarily requires closed, expensive systems.

4.2 Limitations

While this work demonstrates the viability of open-source LLM evaluation, several limitations warrant acknowledgment and consideration for future research.

4.2.1 Limitations of Prometheus 2

Despite strong overall performance, Prometheus 2 faces several inherent constraints:

- **Performance gap with latest proprietary models:** While Prometheus 2 achieves competitive results with GPT-4, the rapid advancement of proprietary models means that continuous development is necessary to maintain competitiveness.
- **Weight merging sensitivity:** The quality of the unified model depends critically on the selection of merging weights. Our work identified effective weights through systematic search, but this process requires computational resources and domain expertise.
- **Format constraints:** Prometheus 2 is designed for specific evaluation formats (Likert-scale scoring and A/B comparison). Adaptation to other evaluation paradigms would require architectural modifications or additional training.

4.2.2 Limitations of Continual Training Methodology

The benchmark-specific adaptation approach also presents practical challenges:

- **Feedback data requirements:** Continual training requires high-quality, benchmark-specific feedback data. Generating such data through human annotation is expensive and time-consuming, potentially limiting the approach’s scalability to numerous benchmarks.
- **Training costs:** While inference with adapted models is efficient, the training process itself requires GPU resources, though these costs remain substantially lower than ongoing API usage fees.
- **Catastrophic forgetting monitoring:** Although we observed minimal catastrophic forgetting in our experiments, practitioners must validate that adapted models maintain acceptable performance on general evaluation tasks.

These limitations do not diminish the contributions of this work, but rather highlight opportunities for refinement and extension in future research.

4.3 Future Directions

The findings presented in this thesis open several promising avenues for future investigation and development.

4.3.1 Extending to More Benchmarks

The continual training methodology demonstrated on BiGGen-Bench can be extended to create a comprehensive suite of benchmark-specific evaluators:

- **Multi-benchmark sequential training:** Investigate techniques for adapting a single model to multiple benchmarks sequentially, building on recent advances in continual learning that mitigate catastrophic forgetting.
- **Cross-benchmark knowledge transfer:** Explore whether adaptation to one benchmark can improve performance on related benchmarks, potentially identifying clusters of similar evaluation tasks that benefit from shared training data.

4.3.2 Improving Base Evaluator Performance

Advancing the capabilities of the foundational Prometheus 2 model remains a critical priority:

- **Scaling to larger models:** Train 70B+ parameter versions of Prometheus 2 to handle more complex evaluation scenarios and narrow the gap with state-of-the-art proprietary models.
- **Enhanced training data quality:** Systematically improve the Feedback Collection dataset by filtering low-quality examples, balancing task distributions, and incorporating more diverse evaluation criteria.
- **Advanced merging techniques:** Explore more sophisticated model merging approaches beyond simple linear interpolation, such as task arithmetic or TIES-merging, which may yield superior unified models.

4.3.3 Open-Source Evaluator Ecosystem

Beyond technical improvements, fostering a vibrant ecosystem around open-source evaluation is essential:

- **Community contribution framework:** Establish platforms and protocols for the research community to contribute benchmark-specific adaptations, evaluation datasets, and improved merging weights, creating a shared repository of evaluation resources.
- **Standardized evaluation protocols:** Develop and promote standardized methodologies for evaluating evaluators themselves, ensuring consistent and reproducible comparisons across different evaluation systems.
- **Reproducible evaluation pipelines:** Create open-source toolkits that integrate Prometheus 2 with popular LLM frameworks to simplify deployment and encourage adoption.

4.3.4 Theoretical Foundations

Finally, deeper theoretical investigation can strengthen the scientific basis of LLM evaluation:

- **Understanding weight merging:** Develop theoretical frameworks explaining why and when weight merging succeeds in creating unified evaluators, potentially drawing on loss landscape analysis and neural network theory.
- **Continual learning theory:** Formalize the conditions under which continual training preserves general capabilities while improving specific performance, contributing to broader continual learning research.

The work presented in this thesis represents a step toward democratizing high-quality LLM evaluation. By demonstrating that open-source models can provide reliable, transparent, and cost-effective evaluation, we hope to enable broader participation in LLM research and development. The future directions outlined above offer concrete paths for the community to build upon these foundations, ultimately advancing the science and practice of language model evaluation for the benefit of all researchers and practitioners.

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